MT 1003 – Calculus and Analytical Geometry

Homework #6

Absolute Maxima and Minima, and Optimization in Computing

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Instructions:

1. Name the submitted file as `Section_RollNumber_HW6` (for example, `A_26I1234_HW6`).
2. Submit the complete homework as one scanned PDF file.
3. Show the objective function, admissible domain, derivatives, critical numbers, endpoint values or boundary limits, and the final interpretation. Use sign charts where needed.
4. Give exact answers wherever possible and interpret the results of application questions.
5. For Section B, include the Python code, generated graphs or numerical output, and your written observations in the same PDF file.

Section A: Analytical Work

I. Absolute maxima and minima

Reminder. For a continuous function on $[a, b]$, compare its values at both endpoints and at all interior critical numbers. Critical numbers include points where $f' = 0$ or f' does not exist while f is defined. On open or unbounded domains, also examine boundary limits and whether the limiting values are attained.

Q1. Closed-interval extrema.

Find the locations and values of all absolute maxima and minima. Show the candidate points and compare their function values.

(a) $f(x) = x^3 - 3x^2 - 9x + 2$ on $[-2, 5]$.

(b) $f(x) = xe^{-x}$ on $[0, 4]$.

(c) $f(x) = |x - 1| + x^2$ on $[-2, 2]$.

Q2. Open and unbounded domains.

Find all absolute extrema, or explain why they do not exist. Check the relevant boundary limits.

$$\text{(a) } f(x) = x + \frac{4}{x}, \quad x > 0; \qquad \text{(b) } f(x) = \ln x, \quad 0 < x \leq 2.$$

II. Two applications

Q3. Processor allocation.

A simplified runtime model is $T(p) = 600/p + 5p$ seconds, where $1 \leq p \leq 32$ is the number of processor cores.

(a) Treat p as continuous and find the absolute minimum runtime.

(b) Cores must be integers. Compare the two neighbouring integers to determine the best feasible allocation.

Q4. A constant prediction model.

A model predicts the same value c for the observations $2, 4, 5, 9$. Its mean squared error is

$$J(c) = \frac{1}{4}\big[(c-2)^2 + (c-4)^2 + (c-5)^2 + (c-9)^2\big].$$

Find the minimizing $c \in \mathbb{R}$, justify that it is the unique global minimum, and explain its relationship to the observations.

III. Interpreting an optimum

Q5. A minimum outside the feasible interval.

A loss function decreases on $[0, 5]$ and has its unconstrained minimum at $x = 5$. If only $0 \le x \le 3$ is allowed, where is the constrained minimum? Explain why solving $f'(x) = 0$ alone can miss it.

Q6. An approached value versus an attained minimum.

Suppose $E(t) > 0$ for every finite $t \ge 0$, while $E(t) \to 0$ as $t \to \infty$. Is 0 an absolute minimum value? Could a positive absolute minimum exist? Explain briefly.

Section B: Optimization with Python

Q7. Check the integer optimum in Q3.

Run the code to compare runtime across the feasible integer allocations.

```
import numpy as np
import matplotlib.pyplot as plt

p = np.arange(1, 33)
T = 600/p + 5*p
best = p[np.argmin(T)]
print("Best integer allocation:", best)
print("Minimum integer runtime:", T.min())

plt.plot(p, T, "o-")
plt.xlabel("Number of cores")
plt.ylabel("Runtime (seconds)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

(a) Compare the result with your exact analysis in Q3.

(b) Restrict the available cores to $1 \le p \le 8$ and rerun the code. Explain the new optimum using the sign of $T'(p)$ on this interval.